

Critical Thinking

CLAIMS & ARGUMENTS II



THE PLAN!

- ☐ Recap

- ☐ What is an Argument?

 - ☐ In-Class Exercises

- ☐ Two Kinds of Reasoning

 - ☐ In-Class Exercises

WHAT IS CRITICAL THINKING?



Last time!

Critical Thinking:

- is *systematic*: it involves distinct procedures and methods.
- entails *evaluation* (used to assess existing beliefs) and *formulation* (used to devise new beliefs).
- operates according to *rational standards*: beliefs are judged by how well they are supported by *reasons*.

WHAT IS CRITICAL THINKING?



Critical Thinking

- We all want to hold *well-supported* beliefs and avoid *unsupported (or false)* ones.
 - Practical reason: beliefs inform our actions.
 - A well-supported belief has *good* reasons supporting it.

WHAT IS CRITICAL THINKING?



- A central (obvious?) tenet:
 - (Tenet) A belief is worth believing if we have *good reasons* to accept it. The better the reasons, the more likely the belief is true, and the more it is worth believing.
 - Another way is to say that *the purpose of thinking critically is to come to correct conclusion.*

WHAT IS CRITICAL THINKING?

- Sounds pretty simple, right?
- The idea: to provide a set of skills and tools to help you think critically
 - Translation: to help you evaluate your own and others' beliefs according to rational standards to determine whether those beliefs are worth holding onto.
 - Or as our text puts it, critical thinking is thinking that critiques. It helps you critique *reasoning*, broadly construed – the thinking used in arriving at decisions, developing plans, coming to conclusions, offering hypotheses, and so forth.

WHY DOES IT MATTER?



- Our lives are defined by our actions and choices, and our actions and choices are defined by our thoughts and beliefs.
 - Why do you choose to eat certain foods?
 - Why do you turn left when you leave the front door?
 - Why do you come to class on time? (At all?)

Last time!

WHY DOES IT MATTER?

- Our lives are defined by our actions and our choices, not by our thoughts



- Why do we eat certain foods?
- Why do we leave the house when we are late?
- Why do we go to class on time? (At a





Last time!

WHY DOES IT MATTER?

- Our lives are defined by our actions and choices, and our actions and choices are defined by our thoughts and beliefs.
 - Why do you choose to eat certain foods?
 - Why do you turn left when you leave the front door?
 - Why do you come to class on time? (At all?)
- So, being critical about your beliefs can help you understand what actions and choices make sense for you.

HOW IT WORKS

- We will begin with some of the *basic concepts* of critical thinking and work out from there.



Last time!

BELIEFS AND CLAIMS, FACT AND OPINION

Belief (or Judgment or Opinion):

- When we come to a conclusion, we have a belief (or a judgment or an opinion).
 - *When a belief (judgment, opinion) is asserted in a declarative sentence, the result is a **claim** or **statement**.*
- **Example:** If you conclude that the battery of your car is dead, you *believe* that the car battery is dead.
 - "The car battery *is* dead."
 - In other words, you make a claim (statement or assertion) that something *is* or *is not* the case.

BELIEFS AND CLAIMS, FACT AND OPINION



Last time!

Statement (or Claim)

- An assertion that something is or is not the case.
- Examples:
 - "Taxes are too high."
 - "The earth is not flat."
 - "Neither of us wants ice cream."
 - "A triangle has three sides."
 - "I am cold."
 - "The capital of Pakistan is Karachi."

BELIEFS AND CLAIMS, FACT AND OPINION

Statement (or Claim)

- Statements can be *true* or *false* (sometimes hard to tell which).
 - Some of the list were false.
- They can be *positive* (asserting a claim: "You are a liar.") or *negative* (denying a claim: "You are not a liar.").

BELIEFS AND CLAIMS, FACT AND OPINION

Last time!

- Beliefs and claims are *propositional*: they can be expressed in true-or-false declarative sentences.

BELIEFS AND CLAIMS, FACT AND OPINION

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - “There is life on Jupiter” is an objective claim.
 - *Why?*
 - Because whether or not life exists there doesn’t depend on whether you (or anyone else) thinks it does.

BELIEFS AND CLAIMS, FACT AND OPINION

Last time!

- An *objective* claim has this characteristic: Whether it is true or false is independent of whether you or anyone else thinks it is true or false.
 - Although *objective* claims are either true or false, we may not know which a given claim is.
 - *Example:*
 - “Portland, Oregon, is closer to the North Pole than the equator” is true.
 - “Portland, Oregon, is closer to the equator than to the North Pole” is false.
 - “More stamp collectors live in Portland, Oregon, than in Portland Maine” is an objective claim whose truth or falsity is not known, at least not by us.

BELIEFS AND CLAIMS, FACT AND OPINION

- Not every declarative sentence expresses an **objective** claim:
 - Judgments, such as “Bruno Mars has swag,” are **subjective**.
 - *Why?*
 - Because whether a **subjective** statement is true or false is *not independent* of whether you think it is true or false.
 - On the contrary, *a **subjective** judgment about something is true if you think it is true (you can't be mistaken).*
 - Because a **subjective** judgment cannot be viewed as probably true or as probably false, it isn't the sort of thing that can be thought of as supportable by evidence.
 - “The strawberry tastes great!”

BELIEFS AND CLAIMS, FACT AND OPINION

Last time!

- In short, **subjective** judgments:
 - cannot be mistaken;
 - are not subject to probability quantifications;
 - are not the sort of thing for which evidence could be given,
 - should not be invoked to dismiss any particular statement as unworthy of discussion.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- Example:
 - If you believe that Portland, Oregon, is closer to the North Pole than to the equator, that opinion happens to be true, and would continue to be true even if you change your mind.

BELIEFS AND CLAIMS, FACT AND OPINION

Last time!

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- We can refer to objective opinions as *factual* opinions or beliefs, but that *does not mean that factual opinions are all true*.
 - Example:
 - "Portland, Oregon, is closer to the equator than to the North Pole" is a factual opinion that is false.

BELIEFS AND CLAIMS, FACT AND OPINION

- Sometimes people talk about the difference between *fact* and *opinion*, having in mind that all *opinions* are *subjective*.
- But some opinions are not *subjective* judgments, because their truth or falsity is independent of what people think.
- We can refer to objective opinions as *factual* opinions or beliefs, but that *does not mean that factual opinions are all true*.
 - A *factual* opinion/belief/claim = an *objective* opinion/belief/claim = an opinion/belief/claim whose truth is independent of whether anyone thinks it is true.

Q4: 4 Would you classify the following claims as objective or subjective?

- a) On a baseball field, the centre of the pitcher's mound is 59 feet from home plate. (Obj.)
- b) Green is the most pleasant colour to look at. (Subj.)
- c) Driving while drowsy is dangerous. (Obj.)
- d) Pit vipers can strike a warm-blooded animal even when it is pitch dark. (Obj.)
- e) Donald Trump is overweight (Obj.)
- f) This lemon tastes great because it isn't bitter. (Subj.)
- g) There is life on Mars. (Obj.)
- h) Your teacher will complain if you text in class. (Obj.)
- i) Your teacher would be crazy not to complain if you text in class. (Subj.)
- j) God exists (Obj.)
- k) Donald Trump is fat (Subj.)

- Read the passage “Determining the Kind of Claim You Should Make” (worksheet)
- *Q5: How might you justify the conceptual claims mentioned in the passage?*

- Read the passage “Determining the Kind of Claim You Should Make” (worksheet)
- *Q5: How might you justify the conceptual claims mentioned in the passage?*
 - 1. For **claims of fact or existence**, you must provide evidence that a situation is, in fact, as you characterize it.
 - 2. **Claims of definition or classification** depend on reasoning about similarities or differences that assigns an entity to some broader class or distinguishes it from other entities.
 - 3. Finally, **claims of cause** or **consequence** connect sets of facts to show that some situation does (or doesn't) follow from or lead to another.
 - 4. **Effective claims of evaluation** or **appraisal** depend on criteria of judgment to justify why something is good or bad (or better or worse than something else)

Q6: *Analyze the claims below.*

1. TV inflates estimates of crime rates.
2. Graphic reports of violence on local TV news leads regular viewers to overestimate by as much as 150 percent both the rate of crime in their neighbourhood and the personal danger to themselves and their families.
3. The Forest Service has spent a great deal of money to prevent forest fires, but there is still a high probability of large, costly ones.
4. "A recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real gross domestic product, real income, employment, industrial production and wholesale-retail sales." (National Bureau of Economic Research, USA)

Q6: *Analyze the claims below.*

1. TV inflates estimates of crime rates.

2. Graphic news coverage of crime on TV leads regular viewers to overestimate the rate of crime in their neighborhoods. ... so vague that we have little idea about what's to come.

3. The FBI spends more money to prevent forest fires, but not on the more costly ones.

4. "A recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real gross domestic product, real income, employment, industrial production and wholesale-retail sales." (National Bureau of Economic Research, USA)

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3. The Federal Government spent a great deal of money to prevent forest

... uses more specific concepts that not only help readers understand the claim more clearly, but also give the writer a richer set of concepts to develop in what follows.

Takeaway Point: The *precision* of the claim signals how the argument is likely to proceed.

USA)

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Takeaway point:

Don't write wordy, long claims for their own sake. But you benefit when you include more terms in early versions of your claims than you ultimately use. The final claim should be only as ***specific*** as your readers need and should include only those concepts that you develop as themes in your argument.

Your readers want you to state your evidence precisely/ They hear warning bells in words that hedge your claim in such a way that they cannot assess its substance.

How much money is a great deal? How probable is a high probability – 30 percent? 80 percent? What counts as large and costly?

neighbors
families.

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Takeaway Points:

Watch for words like *some, most, many, almost, often, usually, frequently, generally*, and so on.

Such words can appropriately limit the breadth of a claim, but they can also fudge it if the researcher didn't work hard enough to get the precise number. What counts as precise, however, differs by field.

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Q6: *Analyze the claims below.*

"As the NBER definition of recession is somewhat **vague**, economists often consider at least two consecutive quarters of decline in U.S. GDP as a simple indicator of a recession. There is no official government recession declaration ...

There is no universal definition of an economic depression, but a depression is a longer, more severe version of a recession." (U.S. News)

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4. "A recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real gross domestic product, real income, employment, industrial production and wholesale-retail sales." (National Bureau of Economic Research, USA)

WHAT IS AN ARGUMENT?

- *Q7: What's the structure of an argument? Give an example.*

WHAT IS AN ARGUMENT?

- **Argument**

- Arguments are the main focus of critical thinking. They are the tool for evaluating the support for our beliefs.

WHAT IS AN ARGUMENT?

- **Argument**

- Definition: a group of statements in which some of them (the *premises*) are intended to support another of them (the *conclusion*).
- Thus, statements can be used in arguments as either premises or conclusions.

WHAT IS AN ARGUMENT?

- Some arguments:
 - 1. Because you want a job that will allow you to make a difference in the world, you should consider working for a charitable organization.
 - 2. When Joseph takes the bus, he's always late. And he's taking the bus today, so I'm sure he's going to be late.
 - 3. A dog would keep me company; so I should get one.
 - 4. My landlord will raise my rent; so I shouldn't get one.

- First, isolate the statements (or “parts”) that make up the argument...
- Statements (Claims) = *an assertion that something is or is not the case*

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Note: this is a one-sentence argument. The sentence contains two statements. One a premise, the other a conclusion.

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- Now, determine the premise(s) and conclusion(s) from among the statements.

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 - 1. Because (premise) you want a job that will allow you to make a difference in the world, (conclusion) you should consider working for a charitable organization.
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 - 3. (premise) A dog would keep me company; so (conclusion) I should get one.
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WHAT IS AN ARGUMENT?

- What distinguishes arguments from all other kinds of discourse?
 - The *logical link* between premises and conclusion
 - The premises *support* the conclusion (*give us reason to believe it*)
- Inference
 - The mental process of reasoning from a premise or premises to a conclusion based on those premises.

WHAT IS AN ARGUMENT?

- It is not always easy to recognize an argument and to locate the premises and conclusion.
- Tip:
 - Are *indicator words* being used?

WHAT IS AN ARGUMENT?

- **Indicator words**
 - Words that frequently accompany arguments and signal that a premise or conclusion is present.

WHAT IS AN ARGUMENT?

- Indicator words
 - Words that frequently accompany arguments and signal that a premise or conclusion is present.
 - Common *premise* indicators:
 - "Because," "in view of the fact," "since," "as indicated by," "given that," "for the reason that," "assuming that," etc.
 - Common *conclusion* indicators:
 - "Therefore," "so," "consequently," "which means that," "as a result," "thus," "it follows that," "ergo," "hence," etc.

WHAT IS AN ARGUMENT?

- Example:
 - 1. [premise] Because you want a job that will allow you to make a difference in the world, [conclusion] you should consider working for a charitable organization.
 - 2. [premise] When Joseph takes the bus, he's always late. [premise] And he's taking the bus today, [conclusion] so I'm sure he's going to be late.

WHAT IS AN ARGUMENT?

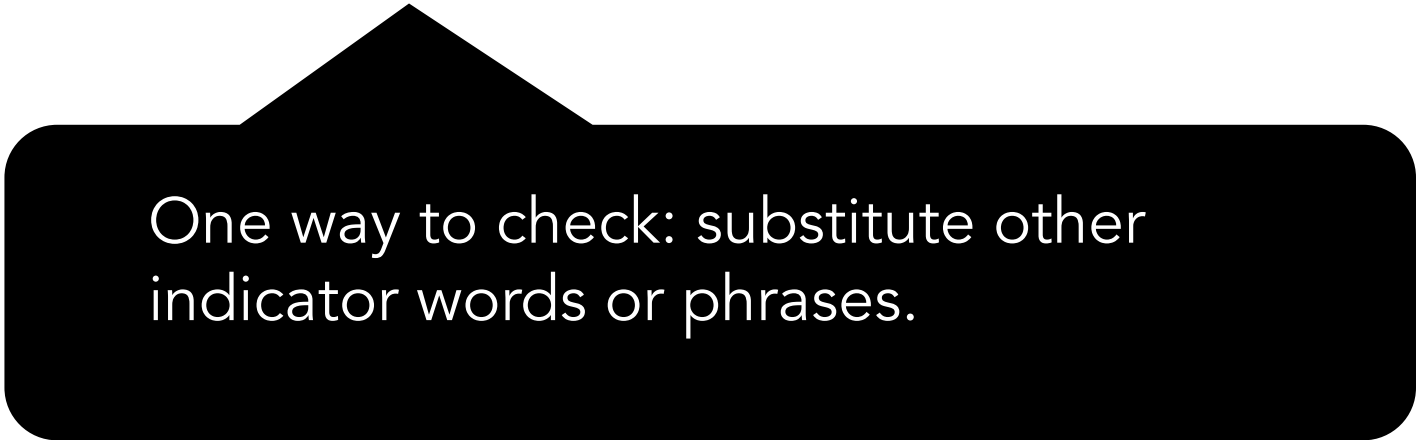
- Examples:
 - 1. [premise] Given that you want a job that will allow you to make a difference in the world, [conclusion] you should consider working for a charitable organization.
 - 2. [premise] When Joseph takes the bus, he's always late. [premise] And he's taking the bus today, [conclusion] thus I'm sure he's going to be late.

WHAT IS AN ARGUMENT?

- But these are rules of thumb:
 - 1. Indicator words can be used in sentences that do not contain arguments.
 - "I am here *because* you asked me to come." [explanation]
 - "I haven't seen you *since* Pakistan Day." [preposition: expresses a relation]
 - "He was so sleepy he fell off his chair." [adverb]

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One way to check: substitute other indicator words or phrases.

WHAT IS AN ARGUMENT?

- But these are rules of thumb:
 - 1. Indicator words can be used in sentences that do not contain arguments.
 - "I am here assuming that you asked me to come." [explanation]
 - "I haven't seen you because Pakistan Day." [preposition: expresses a relation]
 - "He was thus sleepy he fell off his chair." [adverb]

Meaningless or altered meaning =
original words/phrase were not indicating
premises/conclusions.

WHAT IS AN ARGUMENT?

- Q9: *Which of the following passages contain arguments? For any that do, identify the conclusion.*

WHAT IS AN ARGUMENT?

1. *If Elon would read a book, he'd know we can't live on Mars.*

- This is not an argument!
- This is a ***conditional statement***, and has to be treated as one single statement.
 - Conditional statements are made up of two basic parts:
 - The **antecedent** (the if-part) and the **consequent** (the statement that follows the word "then")
 - Note conditionals need not be in the if-then form
- **Conditionals are not arguments** because there is no claim that any statement follows from any part of a conditional statement.

WHAT IS AN ARGUMENT?

2. As a long-time customer, you're already taking advantage of our money management expertise and variety of investment choices. That's a good reason for consolidating your other eligible assets into an IRA with us.

- *Argument. There's a reason being offered for believing the conclusion (opening an IRA (Investment Retirement Account)).*

WHAT IS AN ARGUMENT?

3. *The Directory of Intentional Communities lists more than 200 groups across the country organized around a variety of purposes, including environmentally aware living.*

- Not an argument

WHAT IS AN ARGUMENT?

4. Sweeping changes occurred in demographics, economics, culture, and society during the last quarter of the 20th century. The nation aged, and more of its people gravitated to the Sunbelt. Sprawling “urban corridors” and “edge cities” challenged older central cities as sites for commercial, as well as residential, development. Rapid technological change fueled the growth of globalized industries, restructuring the labor force to fit a “postindustrial” economy.

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- **It's not an argument; it's a *report***
 - It restates and expands on the first statement, rather than giving a concrete reason to believe it actually happened.
 - The purpose of a report is simply to convey information about a subject – e.g., to describe something or narrate a sequence of events (as is the case above).
 - The aim here is to narrate and inform, not to offer reasons why one statement should be accepted on the basis of others.

5. "The most striking thing about Beijing, indeed about all of China, is that there are people everywhere. You need to imagine yourself in a never-ending Macy's sale. There are lines to everything. You have to get in a line to find out which line to be in" (From Cheri Smith, Suttertown News, Sacramento, CA, March 19, 1987).

- It is most probably only a description. It is at least a description because it describes Beijing as being a crowded city containing many lines. Nothing is explained. There is no explanation of why Beijing has so many people, or why it has so many lines.
- You might try to conceive of the passage as being an argument for the conclusion that Beijing is crowded and has lines, but no reasons are given in defence of this claim.
- It probably would be a mistake to say the passage uses the reason that Beijing has many lines to conclude that it is crowded. This would probably be a mistake, because the comments about lines seem to be there to illustrate or describe in more detail the crowded nature of the city, not to make a case for the claim that the city is crowded.

6. "Electrons, and protons, and some other particles ("charged" particles) exert a long-range force on one another. Like gravitation, this force is inversely proportional to the square of the distance between the particles; but unlike gravitation, this force is attractive or repulsive, depending on what particles are involved. Furthermore, this force is much stronger than the force of gravitation-between an electron and a proton, it is about 2×10^{39} times stronger. This force is called the electric force, and particles that exert these forces are said to carry electric charge." (Ohanian, Classical Electrodynamics)

- The first sentence is a claim of cause/consequence.
- The next two sentences are claims of fact/existence.
- The last sentence is a claim of definition/classification.
- However, there is **no argument** here.
- These claims are meant to be believed by the student on the authority of the author.

7. "I wish that everybody could feel the confidence of being alive in a fairly benign world, because I know that most of the advice and conclusions that scientists – scientists not being very scientific about it – have made concerning our planet and the perilous journey we're on, is all bullshit. Because the place is very good at taking care of itself. We don't have to take care of it. And only by being able to read and understand the articles in places like Science and Nature and Scientific American can you really come to that conclusion. You're surrounded by this total barrage of faulty information that's been driven for economic reasons, information that's been basically made up for purposes of getting grants." (Kary Mullis, 1993 Nobel Laureate in chemistry, Annals of Improbable Research, vol. 5, no. 2, 1999)

- This is an argument.
- However, it is a weak one, since the premises are implausible.

8. Many of the world's greatest philosophers were bachelors. For instance, Descartes, Locke, Hume, and Kant were all unmarried.

- Not an argument; rather an illustration
- Illustrations are intended to provide examples of a claim rather than prove or convincingly support the claim.

9. The sky is blue. That's because sunlight is refracted through the air in such a way that other wavelengths of light are diminished.

- This is not an attempt to convince you that the sky is blue – that's obvious.
- This is an explanation, and an explanation is **not an argument**.

TWO KINDS OF REASONING



REFLECT & DISCUSS

1. On a fundamental level, arguments may be classified into two types. What (types) are those, and how do they differ?

TWO KINDS OF REASONING

- Arguments come in two forms:
 - **Deductive argument**
 - An argument intended to provide logically conclusive support for its conclusion.
 - **Inductive argument**
 - An argument in which the premises are intended to provide probable, not conclusive, support for its conclusion.

TWO KINDS OF REASONING

Deductive argument

- A special type of argument:
 - If the premises are true, the conclusion *must* (of logical necessity) be true as well. (This is guaranteed!)

TWO KINDS OF REASONING

Deductive argument

- A special type of argument:
 - If the premises are true, the conclusion *must* (of logical necessity) be true as well. (This is guaranteed!)
- A deductive argument that succeeds in providing such decisive logical support is said to be *valid*.
- Whereas, a deductive argument that fails to provide support is said to be *invalid*.

TWO KINDS OF REASONING

Deductive argument

- An interesting and useful fact about deductive arguments:
 - Decisive logical support is guaranteed by the logical structure of the argument, rather than its contents.
 - Valid arguments are said to have valid structures or *forms*.

TWO KINDS OF REASONING

Deductive argument

- Deductively valid arguments have a particularly interesting property:
 - they are *truth-preserving*.
 - Definition: a characteristic of a valid deductive argument in which the logical structure guarantees the truth of the conclusion if the premises are true.
 - As in "truth is preserved from premises to conclusion."

TWO KINDS OF REASONING

- Examples of other deductively **valid** arguments:
 - Dorothy is an engineer.
All engineers are good at math.
So Dorothy is good at math.
 - All men are mortal.
Socrates is a man.
Therefore, Socrates is mortal.
- If the premises are true, the conclusion **must** be true.
 - The conclusion follows logically from the premises.

TWO KINDS OF REASONING

- An example of a deductively *invalid* argument:
 - Dorothy is an engineer.
Dorothy is a mother.
So all engineers are mothers.
- The conclusion *does not* follow logically from the premises.
 - Think: the premises can be true and yet the conclusion could be false.
 - Truth is not preserved.

TWO KINDS OF REASONING

- An example of a deductively *invalid* argument:
 - All whales have tails.
All whales have eyes.
Therefore, all whales are mammals.
- The conclusion here is true. ***And*** so are the premises.
 - But, the argument is invalid: the conclusion doesn't ***follow from*** the premises offered.

TWO KINDS OF REASONING

- But there is more to good deductive arguments than having the proper structure...
 - That is good argument $\neq$ valid argument
 - Good deductive arguments have true premises...

TWO KINDS OF REASONING

- But there is more to good deductive arguments than having the proper structure...
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 - Good deductive arguments have true premises...

TWO KINDS OF REASONING

- *Sound* argument
 - A deductively valid argument that has true premises.

TWO KINDS OF REASONING

- So, deductively *valid* arguments can have:
 - false premises and a false conclusion
 - false premises and a true conclusion
 - true premises and a true conclusion (sound too!)
- But, a valid argument ***cannot have*** true premises and a false conclusion.

Valid (but unsound) arguments
False premises, false conclusion:

"All books have minds. All desks are books. Therefore, all desks have minds."

"All restaurants are bars. All bars are places that serve pancakes. Therefore, all restaurants are places that serve pancakes."

Valid (but unsound) arguments
False premises, true conclusion:

"My friend Ariane is a cat. All cats like broccoli. Therefore, my friend Ariane likes broccoli

Valid arguments (and sound)
True premises, true conclusion:

Mike is a human. All humans are mammals. Therefore, Mike is a mammal.

A "good" deductive argument. (Valid structure + true premises)

TWO KINDS OF REASONING

- So, for **deductive** arguments:
 - valid argument = if the premises are true, the conclusion must follow.
 - sound argument = a valid argument in which the premises are true.

TWO KINDS OF REASONING

- There are two types of **deductive arguments**:
 - One type relates to *categorical logic*: arguments that relate to categories.
 - The other types relate to *propositional or truth-functional logic*: arguments that relate to propositions which have been compounded from simpler propositions by means of logical operators, such as "not," "or," "if...then..."

Example of a *categorical argument*:

- All movie stars are wealthy.
No wealthy people fly Economy Class.
Therefore, no movie stars fly Economy Class.

"Movie stars," "wealthy people," and "Economy Class" are *categories*.

Example of a *truth-functional argument*:

- If Jones was in Toledo then Jackson was not in Peoria.
In addition, either Jackson was in Peoria or Smith was in Cleveland.
Therefore, if Smith was not in Cleveland then Jones was not in Toledo.
- These arguments are called 'truth-functional' because the truth or falsity ("truth value") of a compound position proposition (e.g., "If Smith was not in Cleveland then Jones was not in Toledo") derives from the truth or falsity of simpler propositions "Smith was not in Cleveland" and "Jones was in Toledo."

TWO KINDS OF REASONING

- Some arguments are not deductively valid arguments, but the premises do provide **some** support for the conclusions.
 - In other words, the premises give us *some reason to believe in the truth of the conclusion*
 - These we call **inductive arguments**

TWO KINDS OF REASONING

- An example of an **inductive argument**:
 - Most engineers are good at math.
Larry Page (co-founder of Google) is an engineer.
Therefore, Larry is probably good at math.
- Is the truth of the conclusion guaranteed by the truth of the premises?

TWO KINDS OF REASONING

- An example of an **inductive argument**:
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No! Why?

TWO KINDS OF REASONING

- An example of an **inductive argument**:
 - Most engineers are good at math.
Larry Page (co-founder of Google) is an engineer.
Therefore, Larry is probably good at math.
- Is the truth of the conclusion guaranteed by the truth of the premises?

But it does give us reason to think the conclusion is **probably** true.

TWO KINDS OF REASONING

Inductive arguments

- An inductive argument that succeeds in providing probable—but not conclusive—logical support for its conclusion is said to be *strong*.
- An inductive argument that fails to provide such support is said to be *weak*.

TWO KINDS OF REASONING

Inductive arguments

- Unlike a deductive argument, the structure of an inductive argument cannot guarantee that the conclusion is true if the premises are true.
 - But the premises can render the conclusion probable, likely, or worthy of acceptance

TWO KINDS OF REASONING

Inductive arguments

- Inductive arguments are incredibly tricky.
- Many people have been stumped by what kind of support they supply.
 - Take my philosophy of science class!

TWO KINDS OF REASONING

- Inductive arguments are not truth-preserving.
 - The truth of the conclusion cannot be guaranteed by the truth of the premises.

TWO KINDS OF REASONING

- Strong inductive arguments with true premises are said to be *cogent*.
 - Good inductive arguments are cogent.
 - Bad inductive arguments are not cogent.

TWO KINDS OF REASONING

A strong inductive argument:

- It is a proven fact that 99 per cent of men love money more than they love their children.
So, it's likely that the next man I meet will love money more than his children.
- But the premise is false, so not a good argument. Not cogent.

TWO KINDS OF REASONING

- Inductive arguments come in four types:
 1. Generalizing from a Sample
 2. Statistical Syllogism (De-generalizing or Reverse Generalization or Instantiating).
 3. Argument from Analogy
 4. Causal Arguments
 - Variation of this type: Inference to the Best Explanation

1. Generalizing from a Sample

- Def.: An argument in which you inductively conclude that all or most or some percentage of all the members of a population have an attribute because all or most or some percentage of the members of a sample of the population have that attribute.
- Example:
 - I've liked the majority of Professor's Moore's lecture so far; therefore, probably I will like the majority of all of Professor Moore's lectures.

2. Statistical Syllogism:

- Def.: This type of argument is the reverse of generalizing from a sample. In it, you conclude that particular members of a population have an attribute because some high proportion of all the population's members have that attribute.
- Example:
 - Most days that are hot and humid are days with thunderstorms. It is hot and humid today. Therefore, probably there will be a thunderstorm today.

3. Argument from Analogy

- Def.: The argument from analogy states that something has an attribute because a similar thing has that attribute.
- Example:
 - Forcing Americans to buy health insurance for their own good is like forcing them to eat broccoli for their own good. It would be wrong to force people to eat broccoli for their own good; therefore, it would be wrong to force them to buy health insurance for their own good.

4. Causal Argument

- Def.: Causal arguments inductively support a claim that asserts or implies cause-and-effect, or, alternatively, use such a cause-and-effect claim as a premise in an argument to establish that something happened or is the case.
- Examples:
 - The toilet is leaking. It isn't possible for the toilet to leak without damaging the floor. Therefore the floor is damaged.
 - There is water on the floor. The only way water could get on the floor is from a leaking toilet. Therefore, the leaking toilet caused the water on the floor.

4. Causal Argument

- Def.: Causal arguments inductively support a claim that asserts or implies cause-and-effect claim that something is the cause of something else.

Example 1: We focus on an event and derive a conclusion about what its *probable* effect is

- Examples:
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 - There is water on the floor. The only way water could get on the floor is from a leaking toilet. Therefore, the leaking toilet caused the water on the floor.

4. Causal Argument

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- Examples:
 - The toilet is leaking and is damaging the floor. The toilet is leaking without its *probable cause* must have been
 - There is water on the floor. The only way water could get on the floor is from a leaking toilet. Therefore, the leaking toilet caused the water on the floor.

TWO KINDS OF REASONING

- Putting things together:
 - *Valid* deductive arguments: truth of premises guarantees the truth of the conclusion
 - *Sound* deductive arguments: are valid and have true premises
 - *Strong* inductive arguments: truth of premises renders the conclusion probably true
 - *Cogent* inductive arguments: strong inductive arguments that have true premises

TWO KINDS OF REASONING

- A note about conclusions:
 - **Deductive arguments** provide conclusions that are absolute; they are either true or not true.
 - **Inductive arguments** provide conclusions that can vary from weak to extremely strong.

TWO KINDS OF REASONING



- Read the passage “Research Arguments”
- 2. Analyze the arguments on your worksheet